

Motivation

- Global competition for the highly-skilled is a known phenomenon, but intertwined study of scholarly migration and collaboration is rare.
- Findings conflict regarding the direction of the effect between the two (effect from mobility on collaboration, bidirectional effects or inconclusive)
- Adopting a spatial approach by representing mobility and collaboration trajectories using word embedding vectors might help to resolve the paradox.
- We chose a random sample of +10k Scopus authors worldwide based on bibliometric criteria, considering internally/internationally (im)mobile scholars as observation/control groups.

Control and observation groups

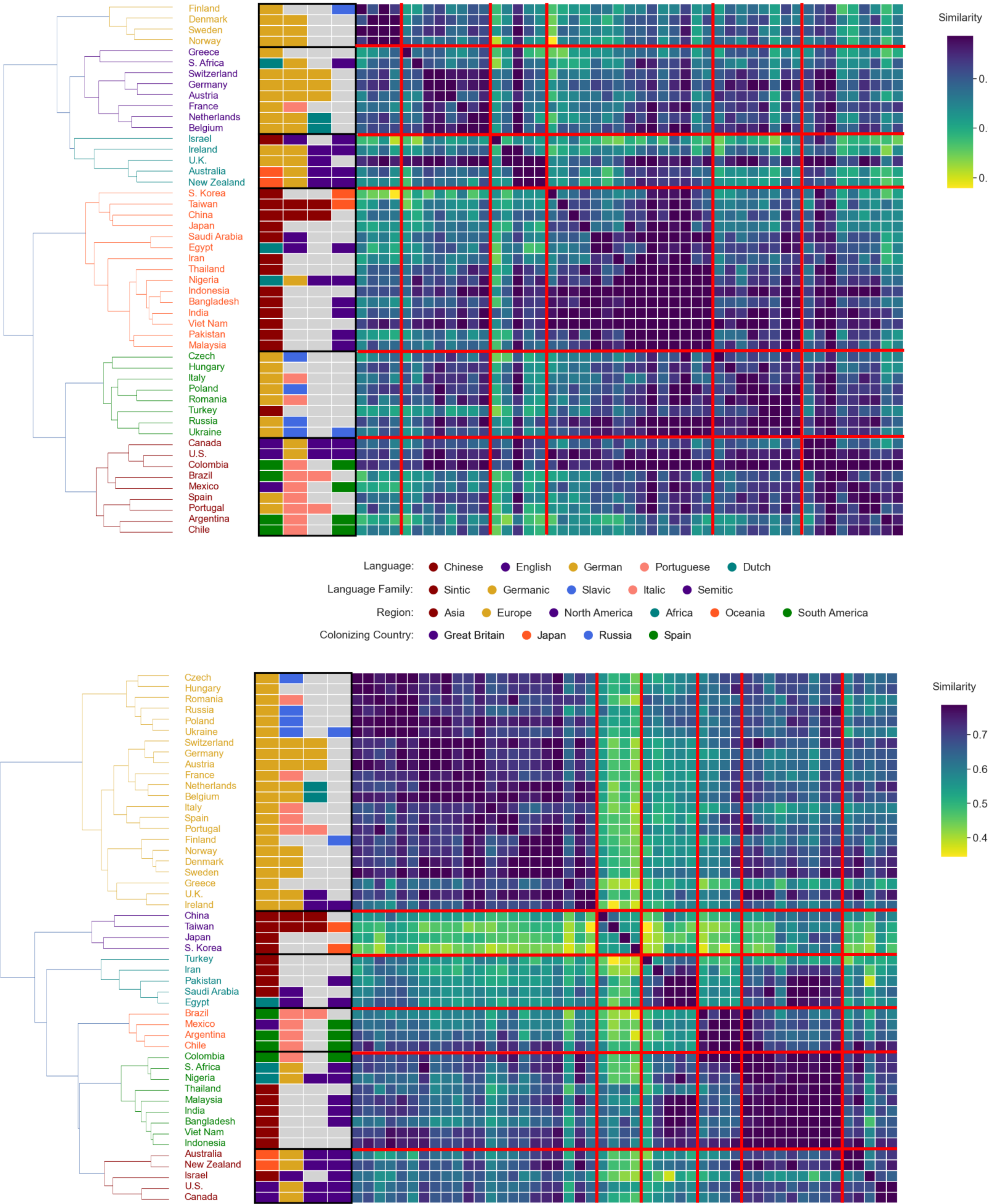
Chosen 10,963 authors (based on 3 bibliometric criteria, N. of publications (o), N. as corresponding author (c), N. in top ranked journals (q1), and their interactions) and the composition of the control (7,399 authors) and observation (3,564) groups.

Group	Indicator combination	Count of authors
Control group	top5-top10% c	1,786
	top5-top10% o	1,867
	top5-top10% oxc	1,050
	top5-top10% oxq1	1,156
	top5-top10% oxq1xc	410
	top5-top10% q1	757
	top5-top10% q1xc	373
Observation group (Potential talents)	top1% c	1,144
	top1% o	712
	top1% oxc	488
	top1% oxq1	649
	top1% oxq1xc	265
	top1% q1	306

Comparing Trajectories

Group	Similarity	# Collabs
non-mobile (2k)	0.93	39.32
mobile internal (3k)	0.94	99.00
mobile international (3k)	0.92	129.12
mobile both (2,963)	0.94	121.00
Control group	0.94	62.71
Potential talents (Obs)	0.92	184.50

Clusters of Migration vs. Collaboration

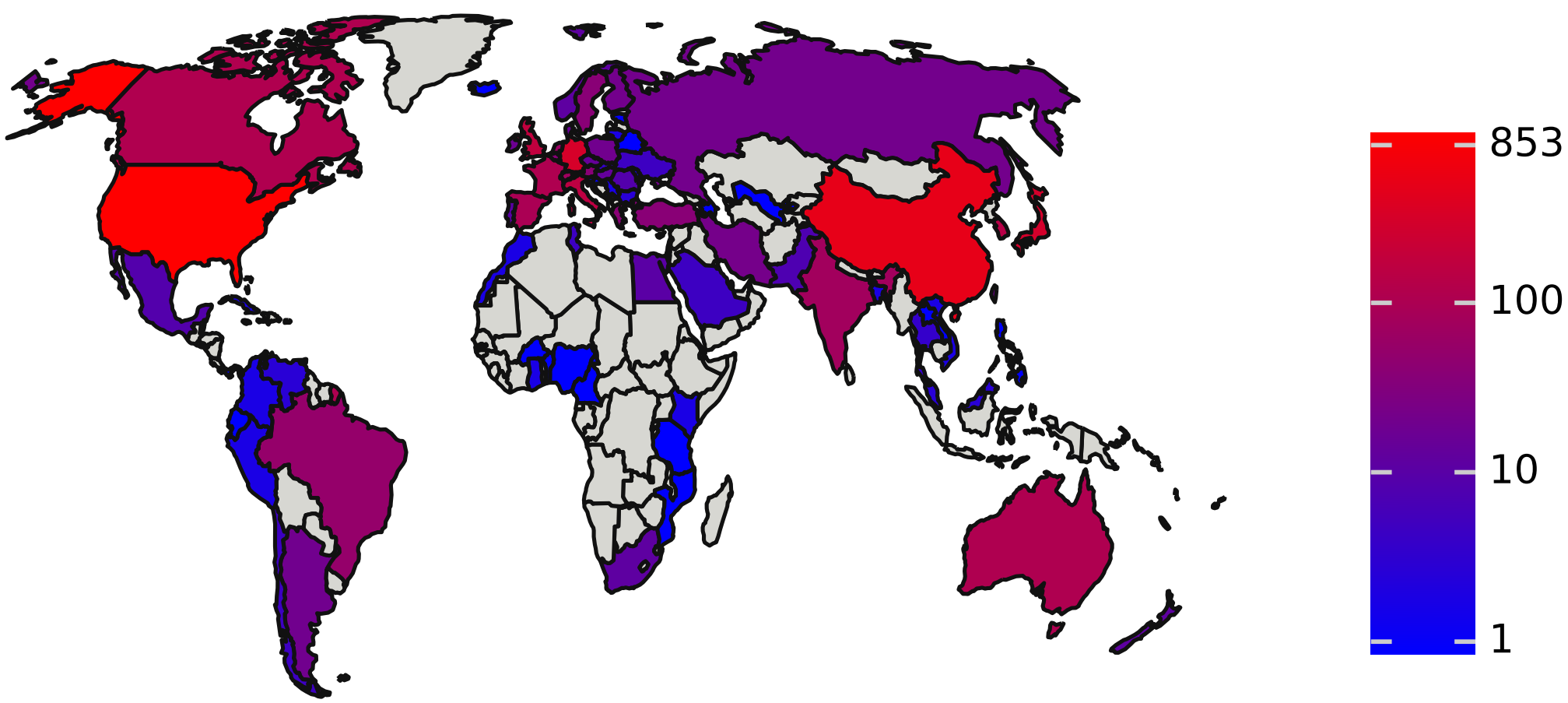


Take away!

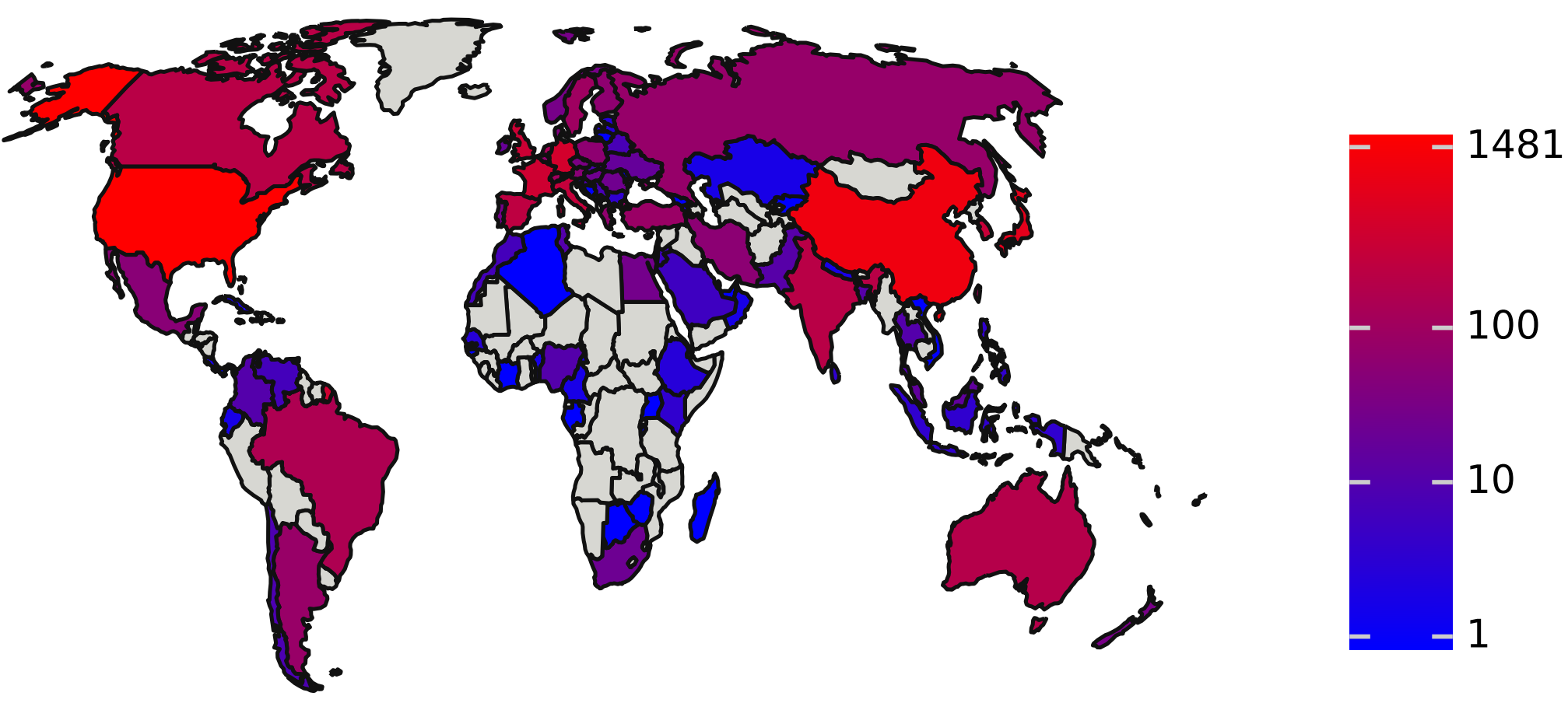
- The literature reports paradoxical results on the mutual effects of scholarly migration and collaboration.
- We adopted the methodological framework proposed by Murray et. al 2021 to represent the trajectory of mobility and collaboration as *word2vec* vectors.
- Results showed that overall collaboration and affiliation trajectories are highly similar. We further observed that mobility is more costly than collaboration which is more diverse.
- Scholars with high levels of collaboration are more likely to be mobile and be among 'Potential talents.'
- We find that most collaborative publications are published before moving to a target institution.
- The average year for the first collaborative publication with the target institution is also further from the movement than the last publication afterwards, suggesting that collaboration is more beneficial as a way to create ties with an institution than as a way to maintain ties after leaving.
- Collaboration with the target institution increases leading up to the move, then decreases following the move.

Control vs. Observation

Number of authors in observation group (log10 scale)



Number of authors in control group (log10 scale)



Selected authors are affiliated with similar countries worldwide (with a few exceptional countries missing from the observation or the control group, e.g., Peru).

Collaboration vs. Migration

Effect of years until move on collaborative publication number (Year of Move as Reference)

